

Introduction to Contract Theory and Auctions
Lecture 9 (mini-lecture)
Moral Hazard: Two applications

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- A risk neutral agent is working for a firm.
- Assume two effort and output levels.
- There is limited liability: $w_0, w_1 \geq 0$
- We have already studied this problem.
- If

$$(p_1 - p_0)(s_1 - s_0) \geq \psi$$

- Firm prefers to induce high effort and the solution is

$$w_0 = 0 \quad w_1 = \frac{\psi}{p_1 - p_0} > 0$$

- Positive wage is called **efficiency wage** because it induces the agent to exert high (efficient level of) effort
- For further development of this idea see **Shapiro and Stiglitz (1984)**

- A risk averse entrepreneur has a project that needs I dollars of investment.
- Return on the project is s_1 or s_0 .
- Entrepreneur's effort is either 0 or 1.
- p_i = probability of s_1 if effort is $i = 0, 1$.
- A risk neutral lender offers a contract that specifies repayment levels conditional on return (r_0, r_1) .

- Assuming that the lender prefers to induce high effort, her problem is

$$\begin{aligned} \max_{r_0, r_1} \quad & p_1 r_1 + (1 - r_0) - l \\ \text{subject to} \quad & p_1 u(s_1 - r_1) + (1 - p_1) u(s_0 - r_0) - \psi \geq 0 \\ & p_1 u(s_1 - r_1) + (1 - p_1) u(s_0 - r_0) - \psi \geq \\ & \quad p_0 u(s_1 - r_1) + (1 - p_0) u(s_0 - r_0) \end{aligned}$$

- Let $w_0 = s_0 - r_0$ and $w_1 = s_1 - r_1$
- The problem is identical to what we have done before.
- At the solution

$$\begin{aligned} r_1 &= s_1 - h \left(\psi + \psi \frac{1 - p_1}{p_1 - p_0} \right) < s_1 \\ r_0 &= s_0 - h \left(\psi - \psi \frac{p_1}{p_1 - p_0} \right) > s_0 \end{aligned}$$

- Lender's expected payoff is

$$p_1 s_1 + (1 - p_1) s_0 - C - I$$

where

$$C = p_1 h \left(\psi + \psi \frac{1 - p_1}{p_1 - p_0} \right) + (1 - p_1) h \left(\psi - \psi \frac{p_1}{p_1 - p_0} \right)$$

is the **cost (to the lender) of moral hazard**.

- The project will be funded only if

$$I \leq \underline{I} = p_1 s_1 + (1 - p_1) s_0 - C$$

- On the other hand, under complete information the project will be funded as long as

$$I \leq \bar{I} = p_1 s_1 + (1 - p_1) s_0$$

- Moral hazard creates **credit rationing** for projects with investment requirement in $[\underline{I}, \bar{I}]$

- If there is limited liability: $r_i \leq s_i$, $i = 0, 1$
- Solution has

$$r_0 = s_0 \quad \text{and} \quad r_1 = s_1 - h \left(\frac{\psi}{p_1 - p_0} \right)$$

- This looks like a **debt contract**
 - In the bad state no money is left to the borrower.
 - In the good state a positive amount is left.